CS331 – Database Systems

Assignment #2 – SQL Queries

Consider the following relational database schema for a movies database:

Movie(movieID, title, year, length, studioID, rating)

Actor(personID, name, gender)

StarsIn(personID, movieID)

Studio(studioID, name, city)

Do the following in SQLiteStudio (10% each)

1. Create the database. Specify the relevant constraints, keys and foreign keys, for each relation.
2. Using the IMDb Website, populate the Movie relation with at least 20 of your favorite movies and the related data in the other relations. It is not necessary to have a complete list of *all* actors in the movies. Do it such that no SQL query below has an empty result. It is not necessary to use the IMDb identifiers. You can find the studio that produced the movie listed on IMDb as "Company Credits" under "Cast and Crew". If there are multiple studios, it suffices to list just one studio.
3. List the movies produced by 'MGM Studios' (use this spelling) after 1990 whose length was more than 90 minutes long, in decreasing order of length.
4. List all the movies by length, shortest first.
5. List the names of all female (gender='F') movie actors in alphabetical order.
6. List the names of studio(s) that produced the Star Wars movies in alphabetical order.
7. List all the movies in which Leonardo DiCaprio starred. Name the movies, the producing studios and their ratings, ordered by year.
8. List the total of movie lengths produced by each studio, in decreasing order.
9. List all studios located in Los Angeles, in alphabetical order.
10. List the average movie rating for each actor, in decreasing order of rating.

Verify that your SQL queries return the correct results.

Submission

This assignment MUST be done in pairs. Please submit the following:

1. Your database as MOVIES.db in SQLite database format, so that ".open movies.db" will load the database in SQLite.
2. Your SQL queries as MOVIES_queries.sql in text format, with a line containing (only) the text ".www" before each query, so that ".read MOVIES_queries.sql" will run the queries in SQLite and display the results in separate browser tabs.

Mark the names of both students CLEARLY in an SQL comment (delimited by /* … */) at the beginning of the MOVIES_queries.sql file. Also cite there any resources (including AI) that you used in your solution.

Grades

This assignment will be graded out of 100%. The weight of this assignment of the total course grade is 3/100. Late submissions will incur a penalty of 2% per day. Submissions will not be accepted at all one week after the official submission date.